

Multiple Choice (10 marks)

1. If the speed of an airplane is measured by means of a stagnation tube, what correction must be applied to the constant-density indication when the plane is cruising at 300 mph?
 - (a) - 10 mph
 - (b) 25 mph
 - (c) -30 mph
 - (d) -15 mph
 - (e) - 5 mph
2. Determine the dew point of a mixture of benzene and n-pentane gases. The composition is 60 mole percent benzene and 40 mole percent n-pentane. The total pressure is 1 atm.
 - (a) 100
 - (a) 75
 - (a) 80
 - (a) 70
 - (a) 60
3. Given that the joint density function of X and Y is $f(x, y) = 6x^2y$ where $0 < x < 1$ and $0 < y < 1$, $= 0$ elsewhere, find $\Pr[0 < x < (3/4), (1/3) < y < 2]$.
 - (a) $3/4$
 - (b) $2/3$
 - (c) $1/4$
 - (d) $5/8$
 - (e) $7/8$
4. Stack is also known as
 - (a) FIFO memory
 - (b) Flash memory
 - (c) LIFO memory
 - (d) LILO memory

5. A box contains 7 red, 5 white, and 4 black balls. What is the probability of your drawing at random one red ball? One black ball?

- (a) $P(R) = 2/3, P(B) = 1/3$
- (b) $P(R) = 5/14, P(B) = 4/14$
- (c) $P(R) = 1/2, P(B) = 1/4$
- (d) $P(R) = 7/15, P(B) = 4/15$
- (e) $P(R) = 7/16, P(B) = 1/4$

True / False (10 marks)

Answer True or False

- 6. True or False: The solution for $y[n]$ is given by $y[n] = (27/20) 3^n - (3/5) (1/2)^n + (1/4) (1/3)^n$.
- 7. True or False: The speed of data transmission in a 4-G network typically ranges from 100 mbps to 1 gbps.
- 8. True or False: The coefficient of compressibility for an ideal gas at constant temperature is given by $(nRT) / (pV^2)$.
- 9. True or False: The rocket thrust can be calculated to be 30,000 lb using the given fuel consumption and exhaust velocity.
- 10. True or False: The fatigue strength of the steel shaft to sustain 80,000 cycles is 164.8 MPa.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Develop a recursion equation for $y[n]$, representing the sum of the squares of integers from 0 to n . Discuss its implications and derive the initial condition for $y[0]$.
- 12. Calculate the force exerted on a second negative point charge of -10^{-4} coulombs, located 50 cm from a -10^{-6} coulomb charge at the origin, and discuss the implications of charge interactions.

End of Paper